Macro

Control Volume Balance

$$0 = \frac{dm}{dt} + \iint_{CS} (\hat{n}\cdot\underline{v})\rho\, dS$$

$$\sum_{on\,CV} \underline{f} = \frac{d(m\underline{v})_{CV}}{dt} + \iint_{CS} (\hat{n}\cdot\underline{v})\rho\,\underline{v}\, dS$$

Macro → Micro
- arbitrary CV
- Express terms as integrals
- Leibniz rule & divergence theorem

Microscopic Balance

$$\frac{\partial\rho}{\partial t} + \underline{v}\cdot\nabla\rho = -\rho(\nabla\cdot\underline{v})$$ (Mass)

$$\rho\left(\frac{\partial\underline{v}}{\partial t} + \underline{v}\cdot\nabla\underline{v}\right) = -\nabla p + \nabla\cdot\underline{\underline{\tau}} + \rho\underline{g}$$ (Cauchy momentum)

ρ = density
$\underline{v}$ = velocity
p = pressure
$\underline{g}$ = gravity

+ Newton, incom. → Navier Stokes

1) Choosing CV (can be arbitrary)
↓
writing out equations specific to that CV
↳ Have to rewrite & recalculate if CV is changed

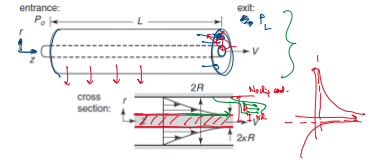
→

2) Already written for an arbitrary, microscopic CV
↳ applicable to any problem ✓
↳ Just need to change boundary conditions & get velocity & pressure profiles!

P → flow direction

PROBLEM: 6.49 Flow Problem: Axial annular drag with pressure drop, wire coating. Repeat Problem 43 with an imposed pressure gradient $-\Delta p/L$ in the flow direction. Calculate the velocity field only.

PROBLEM: 6.43 Flow Problem: Axial annular drag, wire coating. An incompressible Newtonian fluid fills the annular gap between a cylindrical wire of radius kR and an outer shell of inner radius R (Figure 6.22). The wire is pulled to the right at speed V_0. There is no pressure variation throughout the apparatus. Calculate the steady state velocity profile, the flow rate, and the force on the wire. This geometry occurs in wire-coating. Answer: [redacted]



r-direction

$$\rho\left(\frac{\partial v_r}{\partial t} + v_r\frac{\partial v_r}{\partial r} + \frac{v_\theta}{r}\frac{\partial v_r}{\partial\theta} - \frac{v_\theta^2}{r} + v_z\frac{\partial v_r}{\partial z}\right) = -\frac{\partial P}{\partial r} + \rho g_r + \mu\left[\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}(rv_r)\right) + \frac{1}{r^2}\frac{\partial^2 v_r}{\partial\theta^2} - \frac{2}{r^2}\frac{\partial v_\theta}{\partial\theta} + \frac{\partial^2 v_r}{\partial z^2}\right]$$

r_c

$$0 = -\frac{\partial P}{\partial r} + \rho g_r \Rightarrow \frac{\partial P}{\partial r} = 0$$

P, v

θ-direction

$$\rho\left(\frac{\partial v_\theta}{\partial t} + v_r\frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r}\frac{\partial v_\theta}{\partial\theta} + \frac{v_r v_\theta}{r} + v_z\frac{\partial v_\theta}{\partial z}\right) = -\frac{1}{r}\frac{\partial P}{\partial\theta} + \rho g_\theta + \mu\left[\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}(rv_\theta)\right) + \frac{1}{r^2}\frac{\partial^2 v_\theta}{\partial\theta^2} + \frac{2}{r^2}\frac{\partial v_r}{\partial\theta} + \frac{\partial^2 v_\theta}{\partial z^2}\right]$$

$$\frac{\partial P}{\partial\theta} = 0$$

$P = f(z)$ only

$P_L = f(L)$

$$\frac{\partial P}{\partial z} \sim \frac{\Delta P}{\Delta z} \sim \frac{\Delta P}{L}$$

z-direction

$$\rho\left(\frac{\partial v_z}{\partial t} + v_r\frac{\partial v_z}{\partial r} + \frac{v_\theta}{r}\frac{\partial v_z}{\partial\theta} + v_z\frac{\partial v_z}{\partial z}\right) = -\frac{\partial P}{\partial z} + \rho g_z + \mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 v_z}{\partial\theta^2} + \frac{\partial^2 v_z}{\partial z^2}\right]$$

$$0 = -\frac{\partial P}{\partial z} + \mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right)\right]$$

$\frac{\Delta P}{L}$ = constant

Navier Stokes

1) Characterize the flow ✓
$\underline{v}$, $\underline{g}$ = ?
2) Mass / Continuity ✓ Balance
3) Navier Stokes ✓
4) BC's

6.43: No pressure variation ; $\underline{v} = \begin{pmatrix} 0 \\ 0 \\ v_z(r) \end{pmatrix}_{RZ}$; $\underline{g} = \begin{pmatrix} g_r \\ g_\theta \\ 0 \end{pmatrix}_{RZ}$

Mass balance / Continuity equation:

SS, incompr.

$$\frac{\partial\rho}{\partial t} + \underline{v}\cdot\nabla\rho = -\rho(\nabla\cdot\underline{v})$$

$v_r = v_\theta = 0$, v_z

$$(\nabla\cdot\underline{v}) = 0$$ coordinate system

$$\Rightarrow \frac{1}{r}\frac{\partial}{\partial r}(rv_r) + \frac{1}{r}\frac{\partial v_\theta}{\partial\theta} + \frac{\partial v_z}{\partial z} = 0$$

$$\frac{\partial v_z}{\partial z} = 0$$

Newtonian, incomp.

Start w/ Navier stokes:

r-direction : $0 = \rho g_r$

θ-direction : $0 = \rho g_\theta$

z-direction : $\rho\left(v_z\frac{\partial v_z}{\partial z}\right) = \mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right) + \frac{\partial^2 v_z}{\partial z^2}\right]$

$$\mu\left(\frac{1}{r}\right)\frac{\partial}{\partial r}\left[r\frac{\partial v_z}{\partial r}\right] = 0$$ PDE

$$\frac{\partial}{\partial r}\left[r\frac{\partial v_z}{\partial r}\right] = 0$$

$$r\frac{\partial v_z}{\partial r} = c_1$$

$$\frac{\partial v_z}{\partial r} = \frac{c_1}{r}$$

Integrate

$$v_z = c_1\ln r + c_2$$

Solve for velocity profile

Boundary conditions: @ $r = R$, $v_z = 0$

@ $r = kR$, $v_z = V$

always have to be about the variable you need to solve in the PDE! (its value or its derivative's value)

Flow rate → $v = v_z\hat{e}_z$

(its value or its derivatives value)

Flow rate

$$Q = \iint_S (\hat{n}\cdot\underline{v})\,dS$$

$\underline{v} = v_z \hat{e}_z$

$$= \int_0^{2\pi}\int_{kR}^{R} v_z\, r\,dr\,d\theta$$

$v_z = c_1 \ln r + c_2$

easier to calculate derivative of !

$$\int_{kR}^{R} \underbrace{r \ln r}_{u}\,\underbrace{dr}_{dv}$$

Int. by parts.

$$\int u\,dv = uv - \int v\,du \qquad u = \ln r \quad du = \frac{1}{r}dr \qquad v = \frac{r^2}{2} \quad dv = r$$

$$\int d(uv) = \int v\,du + \int u\,dv$$

$$uv = \int v\,du + \int u\,dv$$

$$\int u\,dv = uv - \int v\,du$$

$$\int r \ln r = \frac{r^2}{2}\ln r - \int \frac{r}{2}\,dr$$

Force on wire ?

$$F = \iint_{\text{solid object}} (\hat{n}\cdot\underline{\underline{\Pi}})\,dS \qquad \underline{\underline{\Pi}} = -p\underline{\underline{I}} + \underline{\underline{\tau}}$$

$\hat{e}_r$

1×3 $(1\ 0\ 0)$

3×3 (1×3)

$r = kR$ Constant

$$F = \iint \hat{e}_r \cdot \begin{pmatrix} -p & 0 & \mu\frac{\partial v_z}{\partial r} \\ 0 & -p & 0 \\ \mu\frac{\partial v_z}{\partial r} & 0 & -p \end{pmatrix}_{r\theta z} (r)\,dr\,d\theta$$

$$\underline{\underline{\Pi}} = -p\underline{\underline{I}} + \underline{\underline{\tau}}$$

$$\begin{pmatrix} \tau_{rr} & \tau_{r\theta} & \tau_{rz} \\ \tau_{\theta r} & \tau_{\theta\theta} & \tau_{\theta z} \\ \tau_{zr} & \tau_{z\theta} & \tau_{zz} \end{pmatrix}_{r\theta z} = \mu \begin{pmatrix} 2\frac{\partial v_r}{\partial r} & r\frac{\partial}{\partial r}\left(\frac{v_\theta}{r}\right) + \frac{1}{r}\frac{\partial v_r}{\partial \theta} & \frac{\partial v_r}{\partial z} + \frac{\partial v_z}{\partial r} \\ r\frac{\partial}{\partial r}\left(\frac{v_\theta}{r}\right) + \frac{1}{r}\frac{\partial v_r}{\partial \theta} & 2\left(\frac{1}{r}\frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r}\right) & \frac{1}{r}\frac{\partial v_z}{\partial \theta} + \frac{\partial v_\theta}{\partial z} \\ \frac{\partial v_z}{\partial r} + \frac{\partial v_r}{\partial z} & \frac{1}{r}\frac{\partial v_z}{\partial \theta} + \frac{\partial v_\theta}{\partial z} & 2\frac{\partial v_z}{\partial z} \end{pmatrix}_{r\theta z}$$

(6.49) Pressure variation

$p \rightarrow$ has to be included.

$$F = \iint (\quad)\,kR\,dr\,d\theta$$

$r\theta_z \rightarrow f_\theta, f_r, f_z$

1) $\underline{v} = \begin{pmatrix} 0 \\ 0 \\ v_z \end{pmatrix}_{r\theta z}$; $\underline{g} = \begin{pmatrix} 0 \\ 0 \\ g \end{pmatrix}_{r\theta z}$

2) Mass Balance : $\nabla\cdot\underline{v} = 0 \Rightarrow \frac{\partial v_z}{\partial z} = 0$

3) Navier stokes :

PROBLEM: 6.52 Flow Problem: Upward pressure-driven flow in a tube, power-law non-Newtonian fluid. Repeat problem 6.44 with a power-law, generalized Newtonian fluid with parameters m and n.

PROBLEM: 6.44 Flow Problem: Upward, pressure-driven flow of a Newtonian fluid in a pipe. Calculate the velocity profile and flow rate for pressure-driven flow of an incompressible, Newtonian liquid in a vertical pipe of radius R. The pressure at the bottom entrance to the tube is P_0; at a point a distance L upward the pressure is P_L. Assume that the flow is well developed and at steady state. Do not neglect gravity.

HW problem

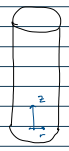
$$\underline{v} = \begin{pmatrix} 0 \\ 0 \\ v_z \end{pmatrix} \qquad \underline{g} = \begin{pmatrix} 0 \\ 0 \\ g_z \end{pmatrix}$$



r-direction

$$\rho\left(\frac{\partial v_r}{\partial t} + v_r\frac{\partial v_r}{\partial r} + \frac{v_\theta}{r}\frac{\partial v_r}{\partial \theta} - \frac{v_\theta^2}{r} + v_z\frac{\partial v_r}{\partial z}\right) = -\frac{\partial P}{\partial r} + \rho g_r + \mu\left[\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}(rv_r)\right) + \frac{1}{r^2}\frac{\partial^2 v_r}{\partial \theta^2} - \frac{2}{r^2}\frac{\partial v_\theta}{\partial \theta} + \frac{\partial^2 v_r}{\partial z^2}\right]$$

$$\underline{v} = \begin{pmatrix} 0 \\ 0 \\ v_z \end{pmatrix} \qquad \underline{g} = \begin{pmatrix} 0 \\ 0 \\ g_z \end{pmatrix}$$



r-direction

$$\rho\left(\frac{\partial v_r}{\partial t} + v_r\frac{\partial v_r}{\partial r} + \frac{v_\theta}{r}\frac{\partial v_r}{\partial \theta} - \frac{v_\theta^2}{r} + v_z\frac{\partial v_r}{\partial z}\right) = -\frac{\partial P}{\partial r} + \rho g_r + \mu\left[\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}(rv_r)\right) + \frac{1}{r^2}\frac{\partial^2 v_r}{\partial \theta^2} - \frac{2}{r^2}\frac{\partial v_\theta}{\partial \theta} + \frac{\partial^2 v_r}{\partial z^2}\right]$$

θ-direction

$$\rho\left(\frac{\partial v_\theta}{\partial t} + v_r\frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r}\frac{\partial v_\theta}{\partial \theta} + \frac{v_r v_\theta}{r} + v_z\frac{\partial v_\theta}{\partial z}\right) = -\frac{1}{r}\frac{\partial P}{\partial \theta} + \rho g_\theta + \mu\left[\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}(rv_\theta)\right) + \frac{1}{r^2}\frac{\partial^2 v_\theta}{\partial \theta^2} + \frac{2}{r^2}\frac{\partial v_r}{\partial \theta} + \frac{\partial^2 v_\theta}{\partial z^2}\right]$$

z-direction

$$\rho\left(\frac{\partial v_z}{\partial t} + v_r\frac{\partial v_z}{\partial r} + \frac{v_\theta}{r}\frac{\partial v_z}{\partial \theta} + v_z\frac{\partial v_z}{\partial z}\right) = -\frac{\partial P}{\partial z} + \rho g_z + \mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 v_z}{\partial \theta^2} + \frac{\partial^2 v_z}{\partial z^2}\right]$$

Power Law : Non-newtonian behavior :

Start from Cauchy balance : $\rho\left(\frac{\partial \underline{v}}{\partial t} + \underline{v}\cdot\nabla\underline{v}\right) = -\nabla p + \nabla\cdot\underline{\underline{\tau}} + \rho\underline{g}$

$$\underline{\underline{\tau}} = \eta(\dot\gamma)\,\underline{\underline{\dot\gamma}}$$

$$= \eta(\dot\gamma)\left[\nabla\underline{v} + (\nabla\underline{v})^T\right] = \mu\begin{pmatrix} 2\frac{\partial v_r}{\partial r} & r\frac{\partial}{\partial r}\left(\frac{v_\theta}{r}\right) + \frac{1}{r}\frac{\partial v_r}{\partial \theta} & \frac{\partial v_r}{\partial z} + \frac{\partial v_z}{\partial r} \\ r\frac{\partial}{\partial r}\left(\frac{v_\theta}{r}\right) + \frac{1}{r}\frac{\partial v_r}{\partial \theta} & 2\left(\frac{1}{r}\frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r}\right) & \frac{1}{r}\frac{\partial v_z}{\partial \theta} + \frac{\partial v_\theta}{\partial z} \\ \frac{\partial v_r}{\partial z} + \frac{\partial v_z}{\partial r} & \frac{1}{r}\frac{\partial v_z}{\partial \theta} + \frac{\partial v_\theta}{\partial z} & 2\frac{\partial v_z}{\partial z} \end{pmatrix}_{r\theta z}$$

where $\dot\gamma = |\underline{\underline{\dot\gamma}}|$

$$= +\sqrt{\frac{1}{2}\left(\begin{matrix}\text{Sum of Squares} \\ \text{of each term}\end{matrix}\right)}$$

$$\eta(\dot\gamma) = m\,\dot\gamma^{n-1} \quad \Big\} \text{ Power law}$$

$$\therefore \tau = m\left(-\frac{\partial v_z}{\partial r}\right)^{n-1}\begin{pmatrix} 0 & 0 & \frac{\partial v_z}{\partial r} \\ 0 & 0 & 0 \\ \frac{\partial v_z}{\partial r} & 0 & 0 \end{pmatrix} = \begin{pmatrix} \tau_{rr} & \tau_{r\theta} & \tau_{rz} \\ \tau_{\theta r} & \tau_{\theta\theta} & \tau_{\theta z} \\ \tau_{zr} & \tau_{z\theta} & \tau_{zz} \end{pmatrix}$$

Momentum equation :

$$\rho\left(\frac{\partial \underline{v}}{\partial t} + \underline{v}\cdot\nabla\underline{v}\right) = -\nabla p + \nabla\cdot\underline{\underline{\tau}} + \rho\underline{g}$$

$$\begin{pmatrix} 0 \\ 0 \\ \frac{\partial}{\partial z} \end{pmatrix}_{r\theta z} \qquad \nabla\cdot\underline{\underline{\tau}} = \begin{pmatrix} \frac{\partial \tau_{rz}}{\partial z} \\ 0 \\ \frac{1}{r}\frac{\partial}{\partial r}(r\tau_{rz}) \end{pmatrix}$$

$$\nabla\cdot\underline{\underline{A}} = \begin{pmatrix} \frac{1}{r}\frac{\partial}{\partial r}(rA_{rr}) + \frac{1}{r}\frac{\partial A_{\theta r}}{\partial \theta} + \frac{\partial A_{zr}}{\partial z} - \frac{A_{\theta\theta}}{r} \\ \frac{1}{r}\frac{\partial}{\partial r}(rA_{r\theta}) + \frac{1}{r}\frac{\partial A_{\theta\theta}}{\partial \theta} + \frac{\partial A_{z\theta}}{\partial z} + \frac{A_{r\theta}}{r} \\ \frac{1}{r}\frac{\partial}{\partial r}(rA_{rz}) + \frac{1}{r}\frac{\partial A_{\theta z}}{\partial \theta} + \frac{\partial A_{zz}}{\partial z} \end{pmatrix}_{rtz} \qquad \text{B.3-10}$$

$$\tau_{rz} = \tau_{zr} = m\left(-\frac{\partial v_z}{\partial r}\right)^{n-1}\left(\frac{\partial v_z}{\partial r}\right) = \frac{(-1)}{(-1)}\,m\left(-\frac{\partial v_z}{\partial r}\right)^{n-1}\left(\frac{\partial v_z}{\partial r}\right)$$

$$= \frac{m}{(-1)}\left(\frac{\partial v_z}{\partial r}\right)^{n-1}\left(-\frac{\partial v_z}{\partial r}\right)^{1}$$

$$\tau_{zr} = -m\left(-\frac{\partial v_z}{\partial r}\right)^{n}$$